SSBMF Status Update: Model Refinement, Validation, and Biological Interpretation (July 15, 2026)

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Headline

The recommended model has **not changed** since the 6/18 meeting — it remains **YFB** ($\eta = (YF)\hat{\beta}$), DeSurv-aligned gene selection, $K = 7$ latent programs, no cohort indicator. What changed is that its numbers were **corrected** and its factor count was **stress-tested**. The scientific conclusion is unchanged and now better supported: the model uses **2 of its 7 programs** for survival prediction, and this is robust, not a tuning artifact. **Mean external C-index = 0.627 across 5 held-out PDAC cohorts.**

What was done since 6/18

- **Objective put on a principled footing (§2 of the full report).** The genomics/survival scale imbalance the lab flagged was addressed; the redundant λ mixing parameter was retired (α already plays its role). Honest finding: for YFB this gave **no performance change** — its genomics half and survival half never share a coordinate needing rebalancing. A separate train/test preprocessing bug (external cohorts were rank-transformed while training was per-platform z-standardized — the reverse) was fixed; **this fix alone** explains the headline moving from 0.636 to **0.627** (a correction, not a regression).
- **Why 7 factors when DeSurv used 3? (§4)** Cross-validation genuinely selects $K = 7$, but only **2 factors are survival-active** ($K_{\text{eff}} = 2$) — matching DeSurv's own ~1 prognostic factor. Two independent optimization strategies (warm-start from a higher-K fit, and a joint (K, α) Bayesian-optimization search) both land on $K_{\text{eff}} = 2$, and a deflation-initialization test analytically ruled out initialization as the cause of an earlier ambiguity; $K \in \{4, 5, 7, 8\}$ are statistically indistinguishable (spread 0.0015). $K = 7$ is kept only for one-step reproducibility; $K = 4/K = 5$ are validated equivalents.
- **Study-specific baseline hazard (§5).** Implemented the requested `+ strata(study)` as an optional stratified Cox partial likelihood. Performance-neutral (0.6267 → 0.6263), so available but off by default.
- **Biological interpretation of the two active programs (§6).** Complete, and confirmed by five independent methods.

Validation across simulated and real data

Simulation — joint model vs. a two-step (factorize-then-Cox) baseline (re-confirmed under the corrected model this cycle):

- *Turn survival signal up from zero:* at zero signal, joint ≈ two-step (both near chance); as signal grows, the joint model pulls ahead and the gap widens, in 3 of 4 data-generating designs. One artificial design (exactly-sparse, symmetric factor loadings) reverses this — traced to a specific, understood factor-collapse failure mode, not a survival-coupling defect.
- *Shared vs. study-specific factors:*

Table 1: Held-out C-index by cross-cohort factor structure (5 seeds).

Scenario	Joint (YFB)	Two-step (EBMF+Cox)
All factors shared	0.871	0.856
Hybrid (some shared, some specific)	0.815	0.717
No shared prognostic factors	0.549	0.550

When no shared prognostic signal exists, joint and two-step are equivalent; in the realistic hybrid case (only some factors prognostic) the joint model leads — as much through **stability** as accuracy: the unsupervised two-step baseline collapses on 2 of 5 seeds (dragging its mean to 0.717), while the joint model is stable across all seeds (mean 0.815) and classifies each factor as shared vs. study-specific with perfect accuracy.

Real data — 5 held-out PDAC cohorts (RNA-seq, microarray, proteomics):

Table 2: External validation with 95% bootstrap CI (mean = 0.627).

Held-out cohort	External C-index	95% CI
PACA_AU_seq	0.657	(0.544, 0.772)
PACA_AU_array	0.648	(0.553, 0.747)
Puleo_array	0.645	(0.599, 0.688)
Dijk	0.634	(0.555, 0.708)
Moffitt_GEO_array	0.549	(0.467, 0.625)

Four of five cohorts clear chance with their CI lower bound above 0.5; **Moffitt microarray does not** (CI 0.467–0.625), confirming quantitatively what was previously only a qualitative flag. **Against a two-step (unsupervised EBMF then Cox) baseline on the same patients, a paired bootstrap finds the recommended model significantly more concordant when pooled across all 5 cohorts: +0.042 (95% CI 0.013–0.071).** Individually only the largest cohort (Puleo, $n = 288$) reaches significance alone; the other four all favor the recommended model but are underpowered on their own. Per-platform z-standardization is essential — most non-per-platform preprocessing collapses the survival coefficients to zero.

Biology of the two active programs

An **adverse** program (basal-like / squamous / MET-EGFR-driven) and a **protective** program (classical / differentiated-epithelial) — the established PDAC subtype-survival axis. Confirmed by five independent methods: gene-set enrichment, PurIST subtype concordance ($\rho = \pm 0.58$), external-cohort hazard ratios (adverse HR>1 in 5/5 cohorts, protective HR<1 in 5/5), and direct gene-list overlap with DeSurv's published programs.

Positioning and open items (for discussion)

- **Positioning:** SBMF is best framed as the **Bayesian counterpart to DeSurv** — same design, data, and scoring, differentiated by empirical-Bayes shrinkage, posterior uncertainty, error-in-variables loadings, and a per-gene noise model. A **matched, same-protocol head-to-head with DeSurv is a pending next step**, not a current superiority claim.
- **Open items worth raising:** (1) factor-stability across seeds not yet characterized; (2) the LB parameterization retains a factor-collapse sensitivity (the recommended YFB model does not); (3) the YFB spike-and-slab prior collapses in cross-validation (normal prior used throughout, and is the right call); (4) requested workflow items (automated commit review, RAG literature database, HPC workflow) and a review of DeSurv's own simulation construction remain on the backlog.

Full detail, figures, and per-item disposition of the 6/18 feedback: `docs/reports/ssbmf_progress_report_07_15_26`.